Tanmay Bhuskute

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Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with 2 years of production experience in Kubernetes, AWS, CI/CD, and Python automation, plus test automation and quality gates. Current research in LLM-powered Android testing.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Go, Java, Bash, JavaScript, TypeScript, C/C++, SQL.

Web & Backend: REST APIs, FastAPI, Flask, Django, Node.js, React, HTML/CSS.

Testing & Databases: Pytest, JUnit, Terratest, Integration Testing, Load Testing, PostgreSQL, MySQL.

Developer Tools: Kubernetes, Docker, Helm, ArgoCD, GitLab CI/CD, Git, Linux, AWS, Azure, Terraform, Ansible, SonarQube.

AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline with a uiautomator2/ADB test automation harness for Android GUI bug reproduction, evaluated against **ViBR** across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Executed **44 production environment upgrades** in 4 months, diagnosing live issues across Linux, EC2, and databases.
- Served as Quality Coach and SonarQube Specialist for the Agile Release Train, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.
- Automated **Teamcenter X** workflows with Python (Boto3), cutting deployment time by **40%**, and maintained GitOps CI/CD pipelines with GitLab CI/CD and ArgoCD for rollback-capable deployments.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** with automated rollback and full environment recovery within 15 minutes.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Developed **Go + Terratest** integration tests validating AWS provisioning, service connectivity, and network security configurations, catching **95%** of misconfigurations pre-production.
- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built full-stack **React and MySQL** application digitizing departmental records and deployed on-premises MySQL with authentication and permissions, cutting document retrieval time by **90%** (5 min to 30 sec).

PROJECTS AND RESEARCH

Spendora | AI Finance Platform [Next.js, TS, Supabase, Gemini, Playwright] Jun 2025

- Implemented Supabase auth and PostgreSQL row-level security, validated by **80 test scenarios** (29 unit, 40 database, 11 E2E).
- Benchmarked LLM financial assistant on **100 queries**: **94%** numerical correctness, **90%** tool-selection accuracy.

PolicyPilot | Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Designed **17-tool MCP server** APIs with Zod schema validation at every data boundary, eliminating a class of runtime type-mismatch bugs across all external tool calls.
- Implemented PDF-to-structured-data pipeline with **Vitest** unit and integration tests and axe-core accessibility coverage.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).